AI Use at Work

Developer Adoption, Perceived Productivity, Trust and Challenges Using Stack Overflow Survey 2025

A PROJECT REPORT

Master of Business Administration Online MBA

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November 2026

Revised empirical draft for supervisor review

Candidate, registration and guide details are retained from the supplied draft for confirmation. Institutional acceptance, signatures and required formatting remain subject to review. The original OECD report is preserved separately; its findings are not pooled with these respondent records.

Declaration and review status

Draft declaration for the candidate to review before signing: This report uses publicly released Stack Overflow Developer Survey 2025 responses for secondary analysis. No primary questionnaire was administered by this project. The scope, exclusions, model specifications and limitations are disclosed. The candidate must confirm the declaration and institutional requirements before submission.

Candidate signature _____ Date _____

Supervisor review

The supervisor is requested to review the revised dataset, objectives, research design, interpretation and report format. This draft does not certify institutional approval or completed supervision.

Guide signature _____ Date _____

Acknowledgements

The project acknowledges Stack Overflow and the survey respondents for the published data and questionnaires. The supplied draft identifies Suraj P G as guide. Naramala Kumar Teja and Keerthana P assisted with the project. Roles and acknowledgements should be confirmed by the candidate and supervisor.

Abstract

This Operations Management project examines AI adoption and developers' reported experiences using public Stack Overflow Developer Survey 2025 microdata. The archive contains 49,191 rows, while the publisher reports 49,009 responses in its published analysis. The reason for the 182-row difference has not been established; the present analysis documents its own cohort and does not claim exact reproduction of the published figures. The scope is employed professional developers: 28,774 records before question-specific and covariate exclusions. Five exploratory binary logistic regressions examine adoption, agent-productivity agreement, debugging frustration, complex-task ratings and trust. Models adjust for grouped work experience, organisation size and grouped country; experience is the exposure in the adoption model. Models of user experiences additionally compare usage frequencies. Robust HC1 standard errors and 95% confidence intervals are reported, with Holm correction for five nominated primary contrasts.

AI adoption model: n=19,773; 20+ versus under 5 years of experience; adjusted odds ratio 0.59, 95% CI 0.53–0.67.

Perceived agent productivity model: n=5,383; Daily versus monthly/infrequent agent use; adjusted odds ratio 8.00, 95% CI 6.77–9.45.

Results describe associations within selected respondents. Voluntary recruitment, missingness, reverse causation and self-reported outcomes prevent causal, national-population or objective financial claims. Operational recommendations concern controlled pilots, human review and measurement of real task time and quality.

Keywords: AI adoption; perceived productivity; developer workflow; trust; logistic regression; Operations Management; Stack Overflow.

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1 Introduction and research scope

Software development is an operational process involving task allocation, cycle time, rework and quality control. AI may support this process while generating verification and debugging work. This study examines adoption and perceived benefits and frictions among employed professional developers, retaining Operations Management as the primary domain and statistical Data Analytics as the methodology.

The scope replaces the earlier OECD sector comparison for this revised report. It does not describe Chennai managers, all Indian employees or employer-level performance. OECD public aggregate analysis remains a separate earlier output. No datasets are pooled and no employer-worker linkage is inferred.

1.1 Objectives and research questions

Objective	Research question and model
1 Describe adoption	Among employed professional respondents, how is experience associated with current AI-tool adoption after adjustment? M1.
2 Examine perceived productivity	Among current agent users, how is agent-use frequency associated with agreement that agents increased productivity? M2.
3 Examine operational friction	Among current AI-tool users, how is frequency associated with selecting time-consuming debugging as a frustration? M3.
4 Examine perceived task capability	Among current AI-tool users who provide a task-capability rating, how is frequency associated with a favourable rating? M4.
5 Examine trust	Among current AI-tool users with a trust rating, how is frequency associated with trusting AI accuracy? M5.

These objectives use questions actually present in this survey. Cost optimisation and managerial decision effectiveness are not measured as final outcomes in this revised study. They should not be presented as tested objectives.

2 Evidence and conceptual framework

Stack Overflow's methodology describes fieldwork from 29 May to 23 June 2025 and recruitment mainly through its own channels. Highly engaged users were more likely to encounter invitations. Its AI results page publishes question wording and descriptive findings on usage, trust, frustrations and agents. These publisher materials establish the measurement context, rather than validating causal effects.

The organising framework relates respondent work context and experience to adoption, then examines usage frequency alongside perceived benefits and frictions among users. Country and employer size are included as contextual adjustments. No causal pathway or mediation mechanism is identified. A favourable experience may encourage frequent use, frequent use may change perceptions, or both may reflect other factors.

Productivity agreement concerns perceived benefit. Debugging frustration concerns a selected problem, not measured rework minutes. Complex-task ratings concern perceived capability, not a task benchmark. Trust concerns confidence in AI output, not verified accuracy. These constructs are kept separate and are not combined into an operational-efficiency score.

The literature component in this revision is limited to the publisher's evidence and statistical documentation. A broader peer-reviewed theoretical review and the institution's preferred citation format remain for supervisor review. No claim of a complete systematic literature review is made.

3 Methodology and measurement

3.1 Source population and unit of analysis

Design: quantitative cross-sectional secondary analysis of publicly released respondent records. The unit is one survey respondent. Inclusion requires MainBranch = I am a developer by profession and Employment = Employed. Freelancers, students, hobbyists and other employment categories are outside the primary cohort. No probability weights are supplied or invented. Country means residence, not employer headquarters. Multiple respondents may share an employer; employer identifiers are unavailable.

Cohort step	Records
Public archive	49,191
Professional developers	37,467
Employed professional cohort	28,774
Current AI-tool users within cohort	16,670
Current AI-agent users within cohort	6,117

The published methodology count is 49,009, which is 182 below the downloaded public archive. The archive SHA-256 is recorded in analysis.json. No rows are arbitrarily deleted to force agreement. Findings refer to the downloaded archive and our explicitly filtered cohorts. The discrepancy remains an unresolved provenance limitation.

3.2 Outcomes and coding

Model	Outcome equals 1	Outcome equals 0
M1	AISelect reports daily weekly or monthly/infrequent use	Either non-use response; planning is not adoption
M2	Strongly or somewhat agree agents increased productivity	Neutral somewhat disagree or strongly disagree
M3	Selects Debugging AI-generated code is more time-consuming	Answers frustration question without selecting this option
M4	Very well or good but not great at complex tasks	Neutral bad or very poor rating
M5	Highly trust or somewhat trust accuracy	Neutral somewhat distrust or highly distrust

M2 reconstructs the productivity item from the five AIAgentImpact agreement columns using exact semicolon-delimited labels. Exactly one rating is required. This avoids treating a blank agreement cell as disagreement. M3 excludes a response selecting both no problems and another option; a missing question is never recoded zero. M4 excludes the combined no use for complex tasks / do not know option. Missing outcomes are excluded model by model. Binary coding simplifies original ordered responses and loses information; zero does not mean a measured absence of benefit. Planning, neutrality and nonresponse remain distinct in source counts.

3.3 Predictors missingness and estimation

M1 compares work-experience categories: under 5 years (reference), 5–9, 10–19 and 20 or more. All models adjust for employer size: under 100 employees (reference), 100–999 and 1,000 or more. Country categories are United States (reference), Germany, India, United Kingdom, France and all other nonmissing countries pooled. These are the five most numerous countries in the full archive, not the OECD seven-country frame. Missing country is excluded rather than pooled. Organisation size I do not know, freelancer-only size, missing size and unavailable experience are excluded from complete-case models. Recorded experience outside 1–60 years is excluded as a covariate under a declared plausibility rule; zero cannot be inferred from blanks because the questionnaire instructed zero-experience respondents to leave it blank.

M2 uses agent frequency; M3–M5 use general AI-tool frequency. Each is categorical with monthly/infrequent as reference and weekly and daily contrasts. Restriction to current users is based on the relevant explicit usage question. No comparison of current agent users with nonusers can identify an agent-adoption effect on a user-only outcome. The primary M2 eligibility follows the agent-use answer even where another question is inconsistent; no unrecorded routing is assumed.

For each model $\log[p/(1-p)]$ = intercept + experience indicators + organisation-size indicators + country indicators + frequency indicators where applicable. Logistic maximum likelihood estimates odds ratios exp(beta). HC1 sandwich standard errors produce Wald 95% confidence intervals and p-values under independent-record assumptions. They do not correct voluntary sampling, common-method bias or unobserved employer clustering. All analyses are exploratory and chosen after inspecting survey variables, not preregistered.

Primary contrasts are 20+ versus under 5 years for M1 and daily versus monthly/infrequent use for M2–M5. Holm adjustment covers these five contrasts only. Other coefficients and the sensitivity fit remain exploratory; the confidence intervals are pointwise, not multiplicity-adjusted. No stepwise significance-based selection is performed. Complete cases assume missingness can be ignored conditional on the model, which is unverified and may bias estimates.

4 Results and validation

4.1 Model sample flow

Model	Eligible	Outcome observed	Covariate exclusions	Final n	Outcome = 1
M1	28,774	20,513	740	19,773	16,108 (81.5%)
M2	6,117	5,538	155	5,383	4,035 (75.0%)
M3	16,670	15,423	476	14,947	7,381 (49.4%)
M4	16,670	15,641	498	15,143	5,402 (35.7%)
M5	16,670	16,473	534	15,939	6,013 (37.7%)

Eligible denominators differ because adoption uses the employed cohort, productivity uses current agent users and the remaining models use current tool users. Outcome-observed counts precede complete-case adjustment. Event percentages above refer only to each final model sample and must not be described as population prevalence. Exclusion counts by covariate are stored in analysis.json and may overlap.

4.2 Primary adjusted associations

Model and comparison	Adjusted OR	95% CI	Holm p
M1 20+ versus under 5 years of experience	0.59	0.53 to 0.67	<0.001
M2 Daily versus monthly/infrequent agent use	8.00	6.77 to 9.45	<0.001
M3 Daily versus monthly/infrequent AI-tool use	0.80	0.73 to 0.88	<0.001
M4 Daily versus monthly/infrequent AI-tool use	5.02	4.41 to 5.72	<0.001
M5 Daily versus monthly/infrequent AI-tool use	5.10	4.53 to 5.73	<0.001

M1 AI adoption: 20+ versus under 5 years of experience is associated with lower odds of the coded outcome, conditional on the included covariates (OR 0.59; 95% CI 0.53–0.67; n=19,773).

M2 Perceived agent productivity: Daily versus monthly/infrequent agent use is associated with higher odds of the coded outcome, conditional on the included covariates (OR 8.00; 95% CI 6.77–9.45; n=5,383).

M3 Debugging frustration: Daily versus monthly/infrequent AI-tool use is associated with lower odds of the coded outcome, conditional on the included covariates (OR 0.80; 95% CI 0.73–0.88; n=14,947).

M4 Complex-task ratings: Daily versus monthly/infrequent AI-tool use is associated with higher odds of the coded outcome, conditional on the included covariates (OR 5.02; 95% CI 4.41–5.72; n=15,143).

M5 Trust in AI accuracy: Daily versus monthly/infrequent AI-tool use is associated with higher odds of the coded outcome, conditional on the included covariates (OR 5.10; 95% CI 4.53–5.73; n=15,939).

For M2, OR 8.00 means approximately eight times the adjusted odds of agreeing that agents increased productivity. It does not mean eight times the productivity, an eightfold probability or a causal improvement. Daily users may self-select because they already perceive benefits. M3 concerns odds of selecting one debugging complaint; it does not establish less debugging time.

4.3 Diagnostics and sensitivity

Model	Converged	Max VIF	Minority outcomes per parameter	In-sample AUC
M1	Yes	1.87	333.2	0.609
M2	Yes	2.04	103.7	0.731
M3	Yes	1.88	567.8	0.593
M4	Yes	1.99	415.5	0.663
M5	Yes	1.89	462.5	0.669

All design matrices have full rank and coefficient estimates are finite. Each GLM fit agrees within 0.000001 with a separately fitted Logit likelihood implementation. No fitting warnings were accepted. Variance inflation factors are below 10 and minority outcomes exceed 10 per estimated parameter. Calibration-bin counts, observed proportions, fitted probabilities, Brier scores and McFadden pseudo-R^2 are saved with the results. These are in-sample diagnostics, not out-of-sample validation; adequate convergence is not evidence that omitted variables or selection bias are absent.

As a coding sensitivity check, M2 was refitted after excluding neutral productivity ratings. The daily versus monthly/infrequent OR was 12.59 (95% CI 9.55–16.62; n=4,505), compared with 8.00 in the primary model. Direction is stable but magnitude depends materially on treatment of neutral responses. This sensitivity addresses coding only, not causality or missing-data bias.

Quality checks found no duplicate ResponseId values and 0 conflicting productivity ratings among eligible agent users. M3 excluded 257 contradictory no-problem selections. 9 employed professional records reported experience above 60 years and were treated as unavailable covariates. No imputation, fabricated participants or reconstructed OECD records were used.

5 Discussion recommendations and limitations

The results connect frequent AI use with more favourable reported capability and trust, and frequent agent use with productivity agreement. These patterns can reflect learning, task suitability, selection and reverse causation. The experience contrast in adoption shows heterogeneity within the respondent cohort, not evidence that experienced employees resist innovation. The debugging association is specific to one reported complaint and should not be generalised to total operational cost or quality.

5.1 Managerial implications

Managers can use the survey patterns to formulate local pilot questions. Begin with bounded tasks, define human-review responsibility and record actual completion time, rework, defects and acceptance quality before and during a pilot. Compare similar tasks and account for task complexity. Provide training in verification, prompt use and handling sensitive information. Assess who benefits and who incurs additional review work. These are recommendations for future evaluation, not interventions tested by this cross-sectional survey.

5.2 Limitations

Voluntary developer recruitment prevents representative inference to all developers, India, Chennai managers or firms. Country pooling masks differences and no employer-level clusters can be identified.

Cross-sectional self-reports do not determine temporal direction. Frequency and positive perceptions may influence each other. Task mix, skill, employer policy and other unmeasured factors can confound associations.

Question-specific nonresponse and complete-case exclusion can select respondents. Missing experience may include zero-experience respondents because of questionnaire instructions. No imputation resolves this ambiguity.

Dichotomising ordered ratings changes interpretation. The sensitivity model illustrates dependence on neutral-response handling. Robust errors do not repair sample-selection bias.

The public-archive count differs from the published count by 182 records. The reason remains unverified. No exact replication claim is made.

The analysis does not measure objective output, monetary savings, causal effects, verified code quality or managerial decision effectiveness. Model diagnostics are internal; no predictive deployment or external validation is claimed.

The revised research scope, academic literature coverage, format and unsigned front matter still require supervisor review.

5.3 Conclusion

The public survey supports a reproducible exploratory study of AI adoption and developer perceptions using genuine respondent records. Five fitted models answer five revised questions. Associations are suitable for discussion of workflow adoption, verification and perceived benefit, with explicit limits on generalisation and causality. OECD approval is not required for this independent dataset; acceptance of the revised dissertation scope remains an academic decision.

6 References

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Open Data Commons. Open Database License 1.0. https://opendatacommons.org/licenses/odbl/1-0/

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statsmodels. Multiple testing correction with multipletests; Holm method. https://www.statsmodels.org/stable/generated/statsmodels.stats.multitest.multipletests.html

Appendices and reproducibility

A Questionnaire and coding

The project reuses the publisher's questionnaire. It does not claim a newly administered or locally validated questionnaire. questionnaire_extract.csv records the exact question wording and source variable names. Outcome coding is specified in Section 3.2; organisation and experience mappings and country categories are explicit in stackoverflow_analysis.py. Full source question files are available from the referenced public archive.

B Output files

File	Purpose
analysis.json	Source hashes cohort counts missingness model diagnostics and sensitivity
primary_results.csv	Five primary contrasts with confidence intervals and raw and Holm p-values
M1 to M5 coefficients.csv	All model coefficients including adjustment variables
M1 to M5 model_rows.csv	Source ResponseId coded outcome and fitted probability for audit
M1 to M5 calibration.csv	In-sample calibration bins
descriptive_counts.csv	Question frequencies with explicit cohort denominators
questionnaire_extract.csv	Publisher question wording for the selected variables

C Rerunning the analysis

Install the versions in requirements-stackoverflow.txt in an isolated Python environment. Download results.csv and schema.csv from the official archive using scripts/download_stackoverflow.py. Run python stackoverflow_analysis.py --data work/stackoverflow_2025/results.csv --schema work/stackoverflow_2025/schema.csv. Run python -m unittest discover -s tests -p test_stackoverflow_analysis.py. To regenerate this revised report run python scripts/build_stackoverflow_report.py --template outputs/stackoverflow2025/AI_Use_at_Work_Regression_Report.docx. The delivered DOCX can itself supply the page styles for a rerun, so the original private draft is not required. Output location: outputs/stackoverflow2025. The analysis rejects changed source hashes until reviewed. Word or another compatible renderer is needed to export the DOCX to PDF.

D Validation and external inputs

Component	Status
Title objectives scope	Revised consistently for employed professional developers; supervisor acceptance pending
Variables and source questionnaire	Mapped to actual public survey fields; no primary survey claimed
Sampling	Voluntary survey and complete-case cohorts explicitly documented

Component	Status
Statistical methods	Five fitted logistic models robust intervals sensitivity and diagnostics complete
Interpretation	Associational self-report limits retained throughout
Institutional submission	Guide approval signatures formatting and literature expectations require confirmation

Source archive SHA-256: 2d1f65308877282edfb4470520eabbc08cb499118432a3dcec6a66c086aa2baa

Source schema SHA-256: 1d24951e04eab46c6f9fecef6ce8e6b32a0a14a3f0eecdcf62f70833a74b3ff8

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